



Multiphysics Simulation of Fluid-Structure Interaction and Coupled Sealing Behaviour of Inflatable Barriers in Underground Tunnels

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Abstract: Flexible inflatable structures play an important role in emergency sealing and protection systems where rapid deployment and adaptive contact with complex boundaries are required. However, the operational performance of inflatable sealing systems is strongly influenced by the coupled interaction among internal pressure evolution, nonlinear membrane deformation, and interface contact behaviour under confined conditions. This study investigates the pressure-deformation-contact coupling behaviour of a Z-fold inflatable barrier during the sealing process of an underground tunnel. A nonlinear finite element model was developed using ABAQUS, in which the Yeoh hyperelastic constitutive model was adopted to describe the large deformation behaviour of the membrane material. The inflation process was simulated using the fluid cavity method, and the interaction between the inflatable membrane and tunnel boundary was defined through contact analysis. The evolution of structural morphology, stress distribution, volume variation, and contact development was systematically analysed to reveal the coupled mechanical response of the sealing system. The results showed that the deployment process consisted of three successive stages, including gravitational descent, inflation-driven expansion, and stable sealing. The complete deployment process was achieved within approximately 30 s. The maximum von Mises stress during inflation was 12.15 MPa, while the stabilized stress decreased to approximately 10.13 MPa, remaining considerably below the material tensile strength of 200 MPa. The airbag volume increased from 0.16 m³ to 5.79 m³, and the final contact area with the tunnel wall reached 4.55 m², corresponding to a sealing coverage rate of 81.3%. The results indicate that the sealing performance of inflatable tunnel barriers is governed by the coupled evolution of pressure loading, nonlinear structural deformation, and boundary contact interaction. This study provides new insights into the multiphysics response mechanism of flexible inflatable systems and offers a numerical basis for the design and optimization of rapid-response sealing devices in underground engineering applications.

Keywords: Multiphysics simulation; Fluid-structure interaction; Inflatable sealing system; Nonlinear membrane mechanics; Contact interaction; Underground engineering

1 Introduction

Flexible inflatable membrane structures have emerged as core functional components for underground emergency isolation systems that demand rapid field deployment and adaptive contact against irregular rock and tunnel inner surfaces. Underground mine fire, gas outburst and toxic gas leakage accidents require instant blocking barriers to enclose disaster zones and prevent hazardous gas diffusion across roadways [1]. Traditional masonry plugging structures suffer from long construction cycles and poor geometric adaptability, while inflatable sealing airbags deliver prominent advantages including fast expansion, remote automatic inflation and flexible fitting to variable tunnel cross-sections, which makes them a dominant technical route for underground emergency protection [2]. Equipped with Programmable Logic Controller (PLC) control modules and wireless remote operation units, inflatable barriers support unmanned rapid sealing under high-risk mine disaster environments, yet their overall working performance is governed by complex multi-field coupling interactions between internal air pressure, large membrane deformation and tunnel boundary contact constraints, which requires in-depth numerical analysis [3].

The comprehensive sealing capacity of underground inflatable devices is jointly determined by three interrelated physical processes: dynamic internal pressure build-up, geometric nonlinear deformation of soft membrane materials,

and contact friction/compression between airbag and surrounding tunnel walls [4]. Clarifying the full-cycle coupled mechanical response during gravity descent and air inflation is critical to improve airtightness and structural safety of such sealing facilities. Previous studies have conducted preliminary experimental tests on automatic airbag plugging prototypes, and recent research adopted three-dimensional particle image velocimetry (3DPIV) optical measurement to capture flue gas flow distribution around inflatable barriers inside simulated coal roadways [5]. These experimental works validate the engineering practicability of inflatable sealing technology, but the complete pressure-deformation-contact coupling mechanism covering the whole deployment cycle remains insufficiently revealed.

The initial folding layout exerts a decisive influence on subsequent inflation morphology and dynamic stability of folded inflatable membranes [6]. Different pre-folding modes produce distinct stress concentration regions during expansion; existing LS-DYNA simulations based on the control volume (CV) method confirm that folded configuration dominates the early deployment phase, and the inflation procedure can be divided into three characteristic deformation stages [7]. In aerospace deployable membrane research, systematic finite element studies targeting Z-fold thin films integrated crease mechanical parameters to analyze folding angle and thickness effects on unfolding kinematics [8]. Comparative analysis between Z-fold and spiral curl-fold layouts for automotive safety airbags further verified that Z-folding delivers smoother deployment with less vibration and uniform stress distribution during inflation [9]. Subsequent tubular Z-fold inflation numerical and experimental tests also confirmed the excellent unfolding performance of this folding form for tunnel plugging airbags.

Finite element numerical simulation serves as the primary research tool to quantify deformation, stress evolution and contact development of inflatable barriers in confined underground spaces [10]. The ABAQUS explicit dynamic framework has been widely adopted to simulate tunnel airbag deformation under external confining loads and verify numerical model reliability [11]. Large-scale underground airbag deployment and dimensional optimization research via finite element method systematically summarized staged stress variation rules during continuous inflation and wall contact formation [12]. Two representative numerical algorithms, CV and arbitrary Lagrangian-Eulerian (ALE), were separately applied to simulate full expansion of large tunnel inflatable plugs, providing reference for model selection in folded membrane inflation simulation [13]. Most current studies only focus on single-stage deformation or isolated mechanical indicators, lacking systematic quantitative analysis of full-cycle multi-field coupling evolution integrating air pressure, nonlinear membrane strain and boundary contact area growth.

Precise constitutive characterization of soft membrane materials is a core prerequisite for accurate inflatable barrier numerical modelling [14]. Airbag membranes experience ultra-large tensile strain exceeding 100% during folding and inflation, leading to strong nonlinear stress-strain relations, hence hyperelastic constitutive equations are universally adopted to reproduce material mechanical behaviour [15]. Among mainstream hyperelastic models, the Yeoh constitutive model exhibits superior fitting accuracy over a wide strain range and achieves stable convergence in airbag inflation simulations [16]. Comparative evaluation of multiple hyperelastic models for rubber-like materials under multi-axial loading conditions proved that Yeoh model outperformed Neo-Hookean and Mooney-Rivlin models in capturing large deformation characteristics [17]. Multi-group comparative tests of filled and unfilled elastomer constitutive parameters, multi-scale silicone rubber calibration and viscoelastic membrane constitutive modelling further provide parameter calibration methods for tunnel airbag membrane finite element models [18, 19].

Apart from inflation deployment analysis, scholars have explored the anti-blast and energy-absorbing performance of multi-cavity inflatable barriers under extreme dynamic impact loads. Airbags with different length-diameter ratios show differentiated axial compression deformation under explosion shock waves, with deformation magnitude gradually attenuating from central region to both ends of the barrier. Impact dynamic tests on sealed inflatable capsules demonstrate that membrane self-deformation can absorb roughly 22% external impact energy.

Despite extensive prior research, systematic investigation on full-process multi-physics coupling behaviour of Z-fold tunnel inflatable barriers from folded initial state to complete tunnel wall fitting is still scarce. Most published literature separately discusses anti-impact buffer performance or segmented deployment characteristics, with rare continuous tracking of pressure-driven deformation, internal stress redistribution and progressive contact expansion throughout the whole sealing cycle. The mutual restriction mechanism between membrane nonlinear mechanical properties and rigid tunnel boundaries under confined underground environments still lacks detailed numerical quantification, which forms the research gap addressed by this paper.

Therefore, this study develops a nonlinear finite element model of a Z-fold inflatable barrier for underground tunnel sealing analysis. The Yeoh hyperelastic constitutive model is employed to describe the nonlinear mechanical behaviour of the membrane material, while the fluid cavity method in ABAQUS is adopted to simulate the inflation process. The coupled evolution of structural morphology, stress distribution, volume variation, and contact area development is systematically analysed to reveal the pressure-deformation-contact interaction mechanism during the sealing process. The findings provide new insights into the multiphysics response characteristics of flexible inflatable systems and offer a numerical reference for the design and optimization of rapid-response sealing devices in underground engineering applications.

2 Multiphysics Modelling of the Inflatable Sealing System

2.1 Model Development

As the airbag and underground tunnel models are relatively simple, they were modelled directly in ABAQUS software. The airbag was modelled using a shell type, membrane elements and a quadrilateral mesh (a four-node, three-dimensional membrane element with reduced integration (M3D4R)), whilst the tunnel was modelled using discrete rigid bodies. The airbag and tunnel models are shown in Figure 1.

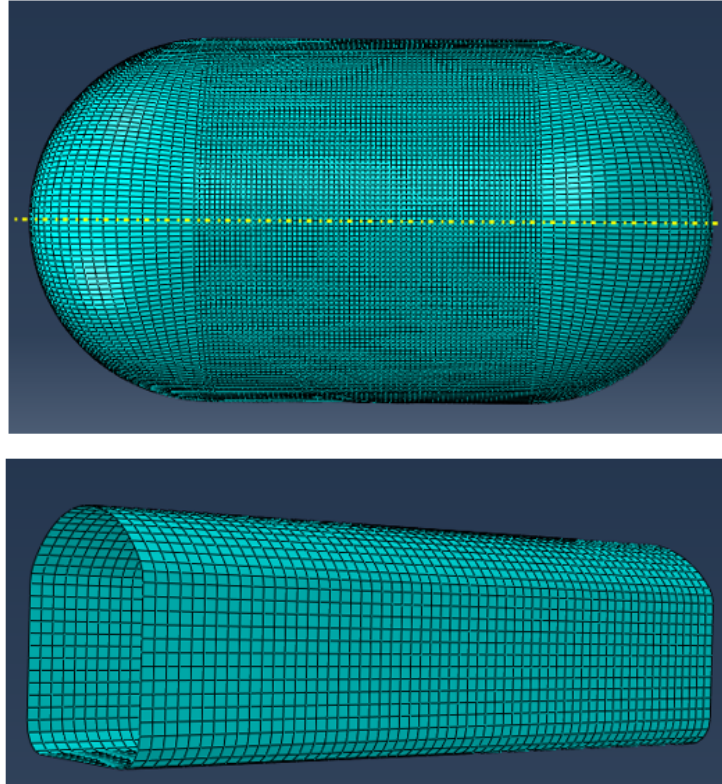


Figure 1. Model of the airbag and tunnel

The airbag membrane undergoes large deformations (typically with strains exceeding 100%) during the folding and inflation processes, and its stress-strain relationship exhibits significant non-linear characteristics. A hyperelastic constitutive model was selected for the airbag material. The software provides several hyperelastic material models; however, as the Yeoh model covers a wide strain range and effectively describes the complex behaviour of large-deformation processes, it yields high-precision results and good convergence. Therefore, the Yeoh model was selected for the airbag material.

2.2 Z-Shaped Folding

This study employed a modified simplified reverse-folding method to generate the initial folded configuration of the underground tunnel airbag. In this method, the unfolded airbag model was first established, and then prescribed constraints and contact conditions were applied to guide the membrane from the unfolded configuration to a flattened folded state. The final deformed configuration was saved and used as the initial state for the subsequent inflation and deployment simulation.

Mesh-independence verification was conducted for the reverse-folding flattening process used to generate the initial folded configuration. In this process, the membrane undergoes large local deformation, self-contact and contact with the prescribed constraints near the fold lines. The maximum stress during the reverse-folding process was used as the mesh-convergence indicator. This stress response mainly reflects the local deformation and self-contact near the fold lines during folded-state generation, and is therefore different from the stress evolution during the subsequent deployment process. As shown in Figure 2, when the mesh size was reduced to 0.03 m, further mesh refinement produced only a limited change in the maximum stress. Considering both computational accuracy and efficiency, an element size of 0.03 m was adopted for the airbag model in the following deployment simulation.

The airbag membrane was folded in a Z-shape along its longitudinal axis, as shown in Figure 3. The main advantage of this folding pattern is that the folded layers can unfold sequentially during inflation, which helps reduce

abrupt deployment, improve the regularity of the unfolding path and avoid severe local deformation concentration. However, the Z-shaped folding process also requires a relatively large initial folding depth, and the mesh near the fold lines should be refined appropriately to ensure numerical stability. This folding configuration is suitable for the inflation and deployment requirements of tunnel inflatable barriers because it can provide a relatively regular unfolding path and stable pressure response during the subsequent deployment process.

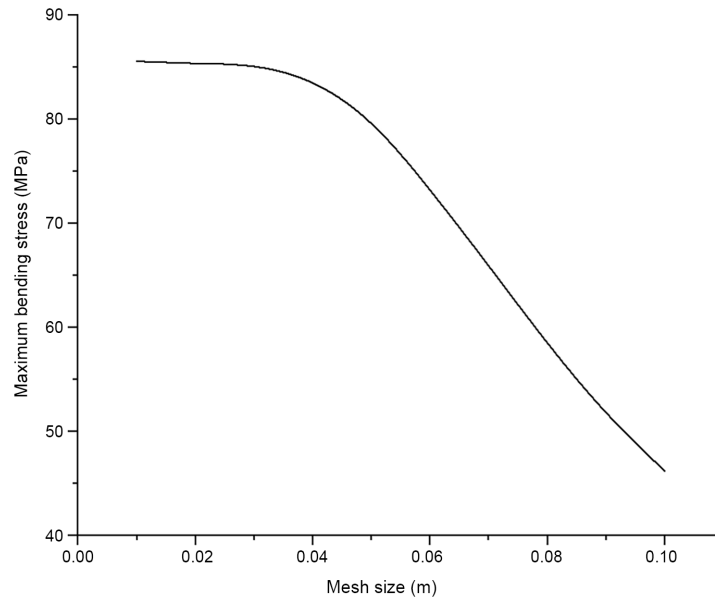


Figure 2. Mesh convergence of the maximum stress during the reverse-folding flattening process

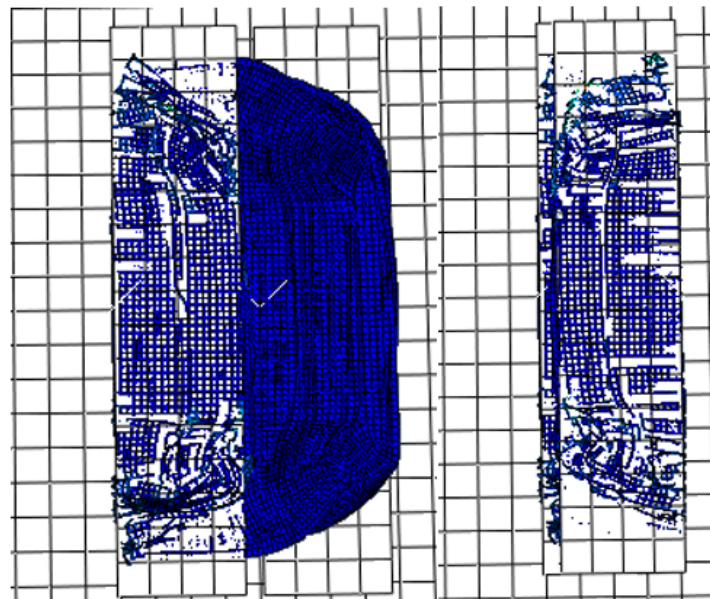


Figure 3. Diagram of the Z-folding process

3 Analysis of the Coupled Sealing Response

First, the Z-folded airbag model is imported into the underground tunnel for assembly. The airbag is secured to the tunnel roof and inflated using a fluid chamber. The assembly diagram is in Figure 4.

The interaction properties between the airbag and the underground tunnel wall are set as: General Contact and Face-to-Face Contact. The fluid cavity was inflated by defining the gas mass input and temperature conditions. This allows for a more realistic simulation of the inflation process. The airbag inflation and deployment process can be

divided into three stages: the gravitational descent stage, the inflation and expansion stage, and the stable sealing stage. A schematic diagram of the process is in Figure 5.

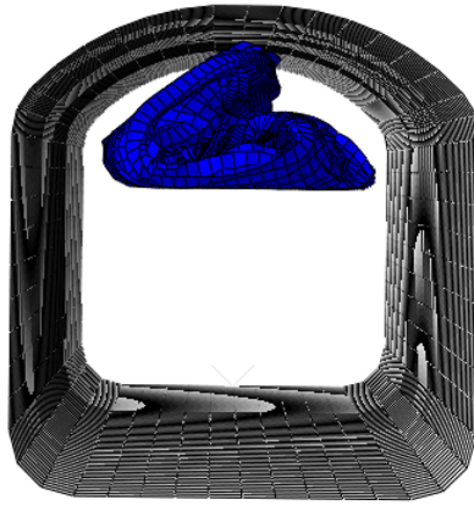


Figure 4. Model assembly diagram

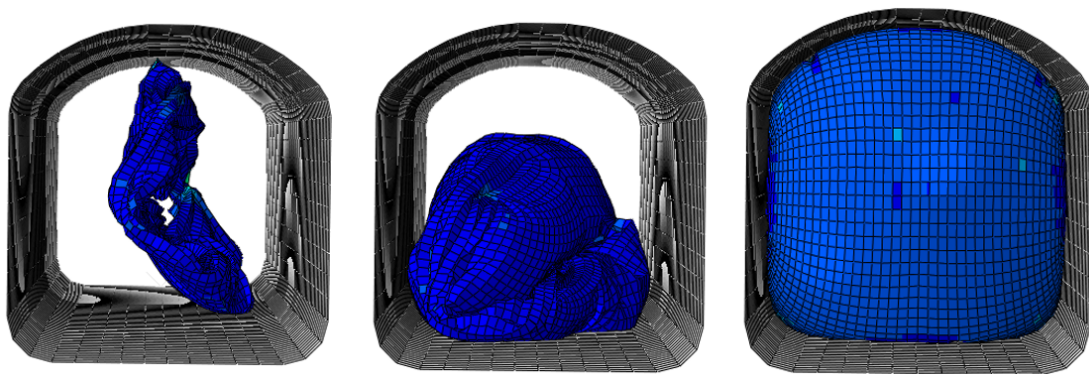


Figure 5. Schematic diagram of the airbag inflation process

Figure 6 shows the von Mises stress contour of the airbag at the final stable sealing stage after inflation and deployment in the underground tunnel. The airbag forms extensive contact with the underground tunnel walls. The stress distribution exhibits the following characteristics: the stress is higher in the contact areas between the airbag and the underground tunnel (roof, side walls and floor), with a peak of 10.13 MPa located in the tunnel corner region; stresses at the free end of the airbag (the end not in contact with the wall) are lower, at approximately 1.6–6.4 MPa; stresses in the airbag membrane are approximately 0.65–3.5 MPa, lower than those in the peak stress zone; and stresses in the area where the base of the airbag contacts the floor, which bears the airbag's own weight and part of the internal pressure, are approximately 3.5–6.4 MPa. The maximum stress in this area will not cause the airbag to rupture or fail.

By extracting the maximum von Mises stress values of the airbag throughout the entire inflation and deployment process, a stress-time curve was plotted, as shown in Figure 7. The figure illustrates that the stress evolution is closely related to the inflation and deployment stages of the airbag.

During the free-fall phase from 0 s to 8 s, whilst the airbag is not yet inflated the internal stresses primarily result from bending deformation in the folded state; the peak stress remains at an extremely low level, approximately 0.2–0.4 MPa. From 8 s onwards, as the lower chamber begins to inflate, the base of the airbag expands and gradually comes into contact with the underground tunnel floor, causing the stress to rise rapidly. By 10 s, the pressure in the lower chamber was approximately 20 kPa, and the maximum von Mises stress in the airbag occurred in the transition zone between the bulging region at the bottom and the folded region, reaching approximately 6.3 MPa. As the lower chamber continued to inflate, the stress level in the main body of the airbag continued to rise; at 18 s, the pressure in the lower chamber reached 80 kPa, at which point both chambers entered the overlapping inflation phase, with the maximum stress in the airbag being approximately 11 MPa.

Between 18 s and 24 s, the inflation of the upper chamber accelerated, and the airbag as a whole transitioned from an asymmetrical shape—narrower at the top and wider at the bottom—to a cylindrical shape conforming to the tunnel roof. During this process, the airbag had to overcome both internal pressure expansion and wall friction to achieve final contact, causing the stress to rise continuously until it reached a peak of approximately 12.15 MPa at 24 seconds. Thereafter, as the pressure in each chamber stabilised, the airbag established comprehensive contact with the tunnel wall, the stress declined slightly and eventually stabilised at around 10.13 MPa. Throughout the entire inflation and deployment process, the peak stress in the airbag was kept below 12.15 MPa, which is far lower than the material's tensile strength (200 MPa) and accounts for only about 6 per cent of the material's strength. This indicates that the airbag possesses a sufficient safety margin throughout the entire inflation and sealing process, and that its structural strength meets the design requirements.

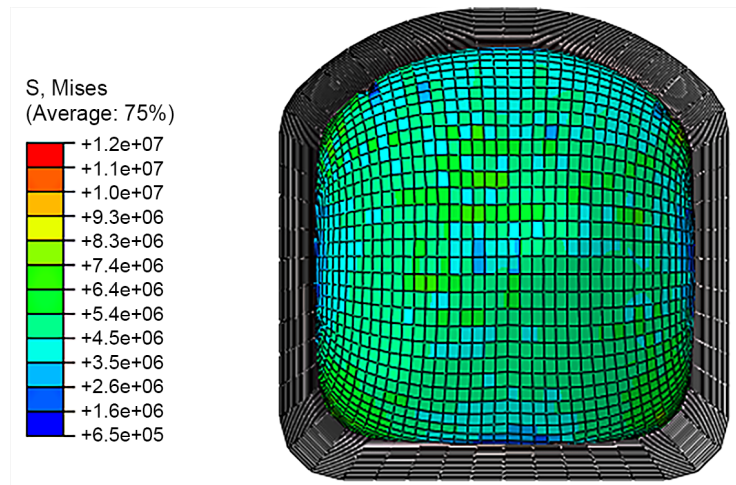


Figure 6. von Mises stress contour plot of the airbag (stable sealing)

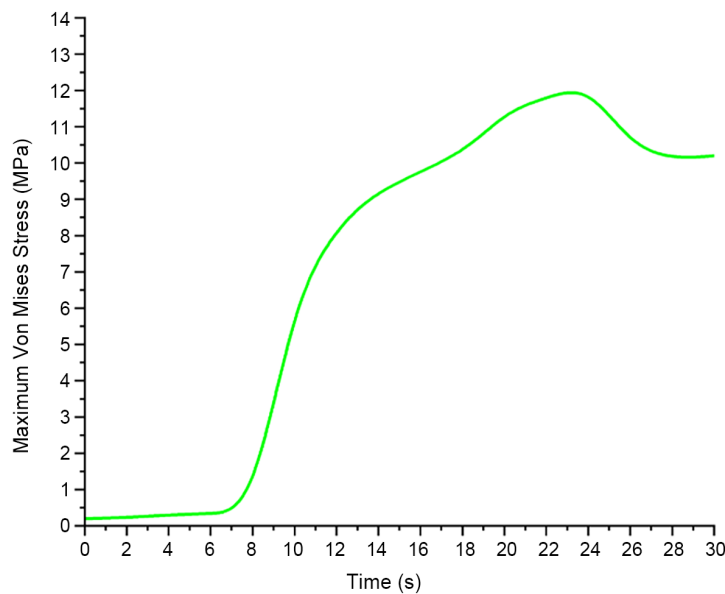


Figure 7. Variation of the maximum von Mises stress of the airbag over time

The volume change curve exhibits three distinct phases. The first phase, from 0 s to 8 s, is the free-fall period, during which the airbag is not yet inflated and its volume remains essentially constant within the range of 0.16 m³ to 0.18 m³, with only minor variations caused by local loosening of the pleats. The second stage, from 8 s to 24 s, is the inflation phase, during which the volume increases in an S-shaped curve. Specifically, from 8 s to 14 s, the lower chamber inflates independently; the volume increases from 0.18 m³ to 1.62 m³ at an average rate of approximately 0.24 m³/s, representing a relatively gentle rate of increase; from 14 s to 20 s, both chambers were inflated simultaneously, with the volume increasing rapidly from 1.62 m³ to 5.15 m³ at an average rate of

approximately 0.59 m³/s, representing a significant improvement in expansion efficiency; From 20 s to 24 s, the final stage of inflation, the rate of volume increase slowed markedly, rising gradually from 5.15 m³ to 5.74 m³, as the airbag progressively achieved final contact with the tunnel wall. The third stage, commencing after 24 s, was the stable sealing period, during which the volume stabilised at around 5.79 m³, with fluctuations of less than 0.1 m³. The marked inflection in the volume growth rate at 14 s coincides closely with the change in inflation strategy. The entire inflation and expansion process lasted approximately 16 s; combined with the 8 s descent phase, the total deployment time was kept within 30 s, meeting the 30 s emergency rescue time requirement. The volume changes during the airbag’s inflation, expansion and sealing process are shown in Figure 8.

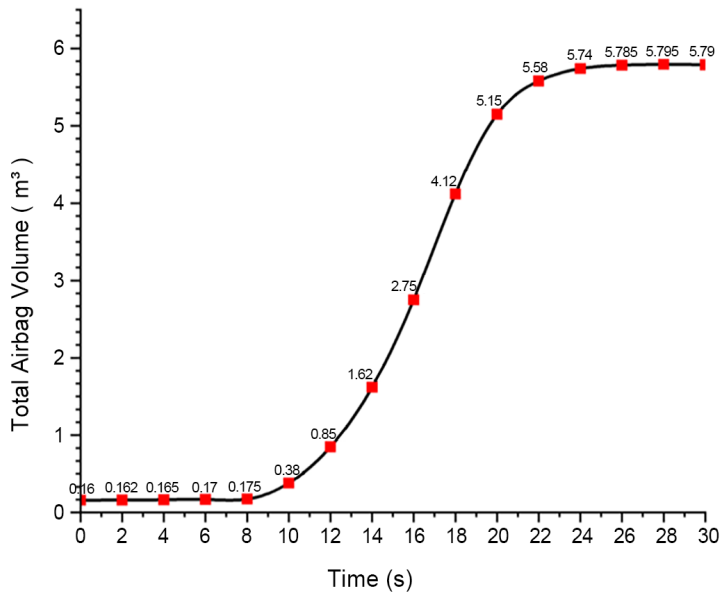


Figure 8. Curve showing the change in total airbag volume over time

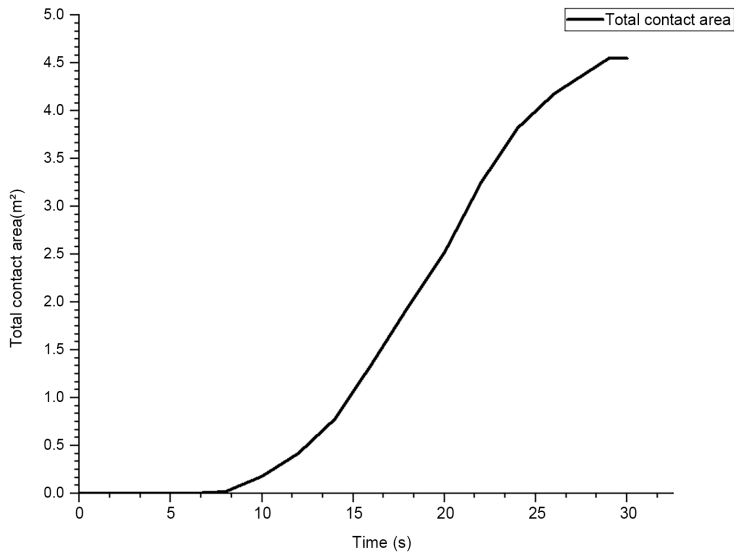


Figure 9. Curve showing the variation of the total contact area between the airbag and the underground tunnel over time

Contact area is a macro-level indicator for assessing the integrity of the seal. This section analyses the evolution of the total contact area between the airbag and the tunnel wall, as well as its constituent parts (roof, side walls, floor and corners).

The total contact area versus time curve is shown in Figure 9. As can be seen from the figure, the evolution of the contact area exhibits distinct phased characteristics: 0–8 s (descent phase): the contact area remains at 0, as the airbag

has not yet made any contact with the tunnel walls. 8–14 s (initial contact phase): the contact area gradually increases from 0 to approximately 0.78 m², corresponding to the process of establishing initial contact between the base of the airbag and the tunnel floor. During this phase, the contact area increased relatively slowly, with an average growth rate of approximately 0.13 m²/s. 14–20 s (rapid expansion phase): The rate of increase in contact area accelerated significantly, reaching 2.52 m² at 20 s, corresponding to an average growth rate of approximately 0.29 m²/s over this 6-second interval. This phase corresponds to the establishment and expansion of contact with the side walls, as well as the onset of contact with the roof. 20–26 s (slowing growth phase): The growth rate gradually decreases, with the contact area reaching 4.18 m² at 26 s. As most of the wall surfaces have been covered, the remaining uncontacted areas are mainly concentrated near corners and free ends, where contact is difficult to establish. 26–30 s (stable phase): The contact area increased slowly to 4.55 m² before stabilising. The theoretical maximum contact area (the area of the tunnel's inner surface within the airbag's coverage zone) was approximately 5.6 m² (calculated as the perimeter of the tunnel cross-section multiplied by the effective length of the airbag, excluding the free end). The actual steady-state contact area was 4.55 m², with a coverage rate of approximately 81.3%. The areas without contact were mainly concentrated in localised sections of the corners and near the free end, with the area without contact at the corners being approximately 0.55 m² and that at the free end approximately 0.50 m².

4 Conclusions

This study developed a nonlinear finite element model of a Z-fold inflatable barrier and investigated its coupled pressure-deformation-contact behaviour during the complete sealing process in an underground tunnel. The numerical framework incorporated the nonlinear mechanical response of the membrane material, pressure-induced nonlinear deformation, and interaction between the inflatable barrier and tunnel boundary.

The results showed that the deployment process consisted of three successive stages, including gravitational descent, inflation-driven expansion, and stable sealing. The complete deployment was achieved within approximately 30 s, satisfying the requirements for rapid emergency sealing. During inflation, the maximum von Mises stress reached 12.15 MPa, while the stabilized stress decreased to approximately 10.13 MPa, which remained considerably below the material tensile strength of 200 MPa. The final contact area between the inflatable barrier and tunnel wall reached 4.55 m², corresponding to a coverage rate of 81.3%, indicating effective sealing performance under confined underground conditions.

The results indicate that the sealing capability of the inflatable barrier is governed by the coupled evolution of internal pressure loading, nonlinear membrane deformation, and boundary contact interaction. This study provides new insights into the multiphysics response mechanism of flexible inflatable sealing systems and offers a numerical basis for the design and optimization of rapid-response sealing devices in underground engineering applications.

Author Contributions

Conceptualization, H.C. and D.X.; methodology, H.C.; software, H.C.; validation, H.C. and D.X.; formal analysis, H.C.; investigation, H.C.; data curation, H.C.; visualization, H.C.; writing—original draft preparation, H.C.; writing—review and editing, H.C. and D.X.; supervision, D.X.; project administration, D.X. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

The authors declare that no generative artificial intelligence or AI-assisted technologies were used to generate research data, conduct numerical simulations, fabricate results, or replace the authors' intellectual contribution. AI-assisted tools were used only for language polishing and manuscript formatting assistance. The authors have reviewed and edited all AI-assisted content and take full responsibility for the accuracy, originality, and integrity of the manuscript.

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